



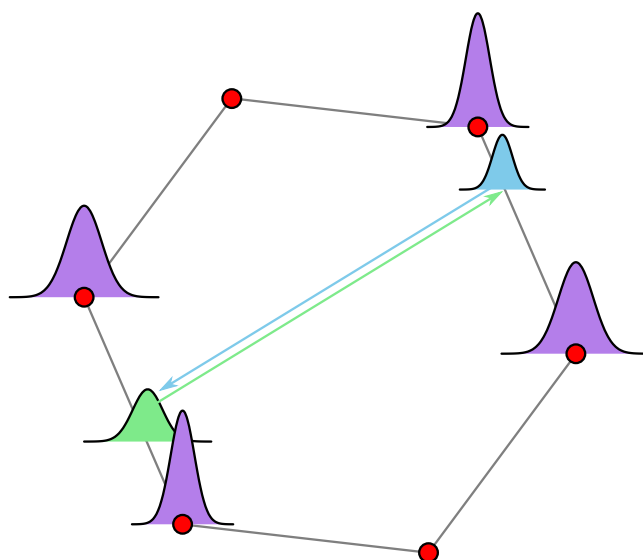
THE UNIVERSITY OF
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FINAL REPORT

The Art of Scientific Computing: Project Title: Quantum Chemistry

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Chapter 1

Introduction

1.1 Graphene

Graphene comprises of a single layer of carbon atoms in a honeycomb lattice. Since its isolation in 2004 (Novoselov et al. 2004), graphene has been studied extensively due to its unique structure and properties. Due to its strength and conductivity, it is an ideal material for high-speed electronic devices, having applications in a range of fields, including electronics, biomedicine, photonics, etc (Singh 2025). Additionally, applications often require confinement into finite structures, such as nanoribbons and nanoflakes.

For an infinite, ideal graphene sheet, the conduction and valence bands meet at the Dirac points, giving a zero band gap. However, for a finite graphene flake, the gap can depend on size, shape, edge structure and symmetries (Jeong and Lee 2024).

- Compare Zig zag and armchair edges

1.2 Hückel Molecular Orbital Theory

- Nearest neighbour β , on-site energy α
- cite original method - huckel
- application in graphene flakes - energy spectra, HOMO-LUMO gaps

Chapter 2

Computational Model

- see appendix for flowchart?
- method for constructing Hamiltonian matrix, slater function, orbital energies, etc
- explain code uses

Chapter 3

Results and Discussion

- ethene results
- benzene results
- compare graphene for varying flake sizes
- compare graphene for band edge types (if have time)

Chapter 4

Conclusion

4.1 Further Work

- limitations and improvements - future work and uses

4.2 Conclusion

- summary of results, validity when compared with literature

Bibliography

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